

Introduction to EE — Module Outline

Syllabus extract: modules only

Module 1: Circuits

- What is voltage and how to measure voltage in series and parallel
- Resistance; formula: $R = \rho \cdot L/A$
- Ohm's law: $V = I \cdot R$
- Write code to blink an LED every 10 ms
- Circuit analysis: apply KCL and Ohm's law to write a system of equations
- KCL and KVL
- Resistors in series and parallel
- Voltage divider
- Current divider
- How to measure voltage, current, and resistance
- Resistive touch panel
- Thevenin equivalent circuit: find V_{th} and R_{th}
- Norton equivalent circuit: find I_n and R_n
- Capacitors in series and parallel
- Inductors in series and parallel
- Source transformation (voltage $\leftrightarrow$ current), worked examples in class

Module 2: Tunable Resistors & Semiconductors

- Tunable resistor (potentiometer) and applications
- Three material types: metals, insulators, semiconductors
- Why silicon; how to obtain p-type and n-type semiconductors
- Energy bands: valence band, band gap, conduction band; carrier movement
- PN diode: forward bias, reverse bias; ideal I–V curve
- Diode equation; calculations using the ideal diode equation
- What is a transistor; MOSFET cross-section and terminal roles
- NMOS and PMOS; how a channel is created when the transistor turns on
- From plots: identify linear and saturation regions; determine off/linear/sat from biases
- CMOS inverter: draw and explain logic inversion operation
- Direct vs indirect semiconductors; which is used for optoelectronics
- LEDs: why different colors have different turn-on voltages
- LED biasing circuit: source + current-limiting resistor
- Photodiode: depletion-region absorption, photocurrent generation; why reverse bias is used
- LED and photodiode spectra; choosing pairs to increase light capture probability

Module 3: Sensors

- What is a sensor?
- How to calibrate a sensor
- Thermistor: find unknown temperature from the thermistor equation
- Solve Wheatstone bridge equations
- Sensitivity = change in output / change in input
- Thermistor readout (voltage divider): derive R_{therm} from V_{out}
- Five parts of a sensor system
- Operational amplifiers (op-amps): why we use them
- Differential measurement (common-mode vs differential signals)
- Inverting op-amp
- Non-inverting op-amp

Module 3: Sensors (cont.)

- Solving op-amp circuit equations (ideal op-amp rules)
- Photodiode readout circuit with an op-amp (transimpedance amplifier concept)
- What is pulse oximetry?
- Why two colors of light in oximetry
- Beer–Lambert law; solve problems
- Oximeter circuits: LED driver (LED + source + current-limiting R) and photodiode readout (PD + op-amp + R)
- Compute oxygenation: $ROS = (Red_AC/Red_DC)/(NIR_AC/NIR_DC)$; $SpO_2 \approx 110 - 25 \cdot ROS$
- What is the Fourier transform?
- Filters: low-pass, high-pass, band-pass, and notch filters

Module 4: Neural Networks

- What are features and why do we use them?
- Workflow: collect data → extract features → train model → deploy model
- Holdout method: training, validation, testing
- Confusion matrix (classification)
- True Positive Rate (Sensitivity)
- True Negative Rate (Selectivity)
- F1 score; relation to dataset balance
- Underfit, good fit, overfit models
- What is a neuron? $y = f(w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b)$
- Activation functions: sigmoid, ReLU, tanh
- Perceptron
- Perceptrons for logic functions: AND, OR, NAND, NOR
- Designing neural network topology
- Example: for an $n \times n$ sensor array, design a network to classify 10 handwritten digits
- Cost function and gradient descent

Module 5: Communications

- What is bitrate and how to calculate bitrate
- Draw an optical fiber; explain how and why light travels
- Noises in communication systems; optical communication system block diagram
- Thermal noise and shot noise
- How to add noises (noise combination)
- Worked examples of calculating noise
- What is bandwidth? What is a filter?
- How filters can improve signal-to-noise ratio (SNR)